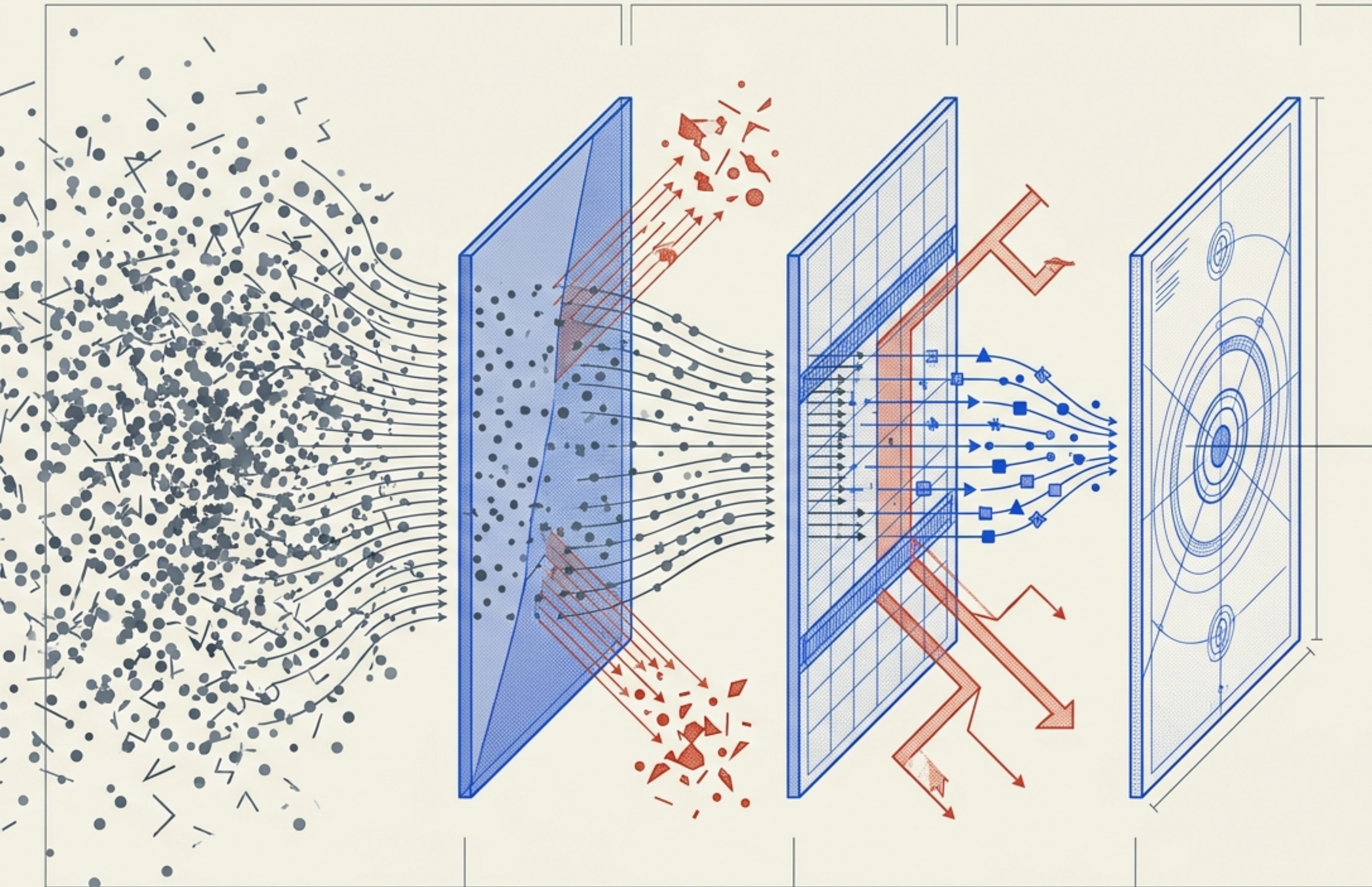


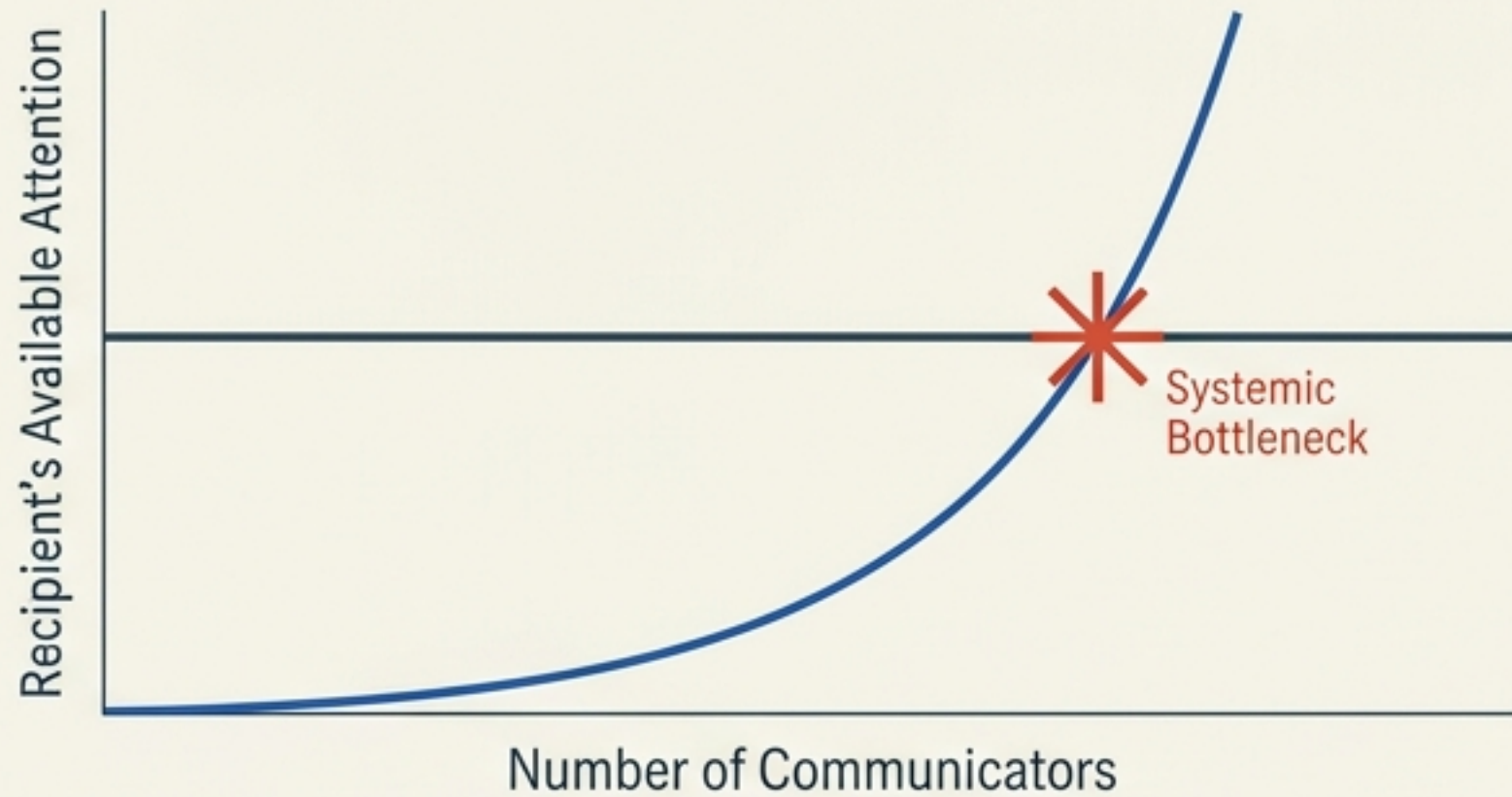


THE ATTENTION COMPILER

CONTENT ROUTING,
REPRESENTATIONAL
SIMPLICITY, AND THE
TRIAGE OF NOVEL
IDEAS.

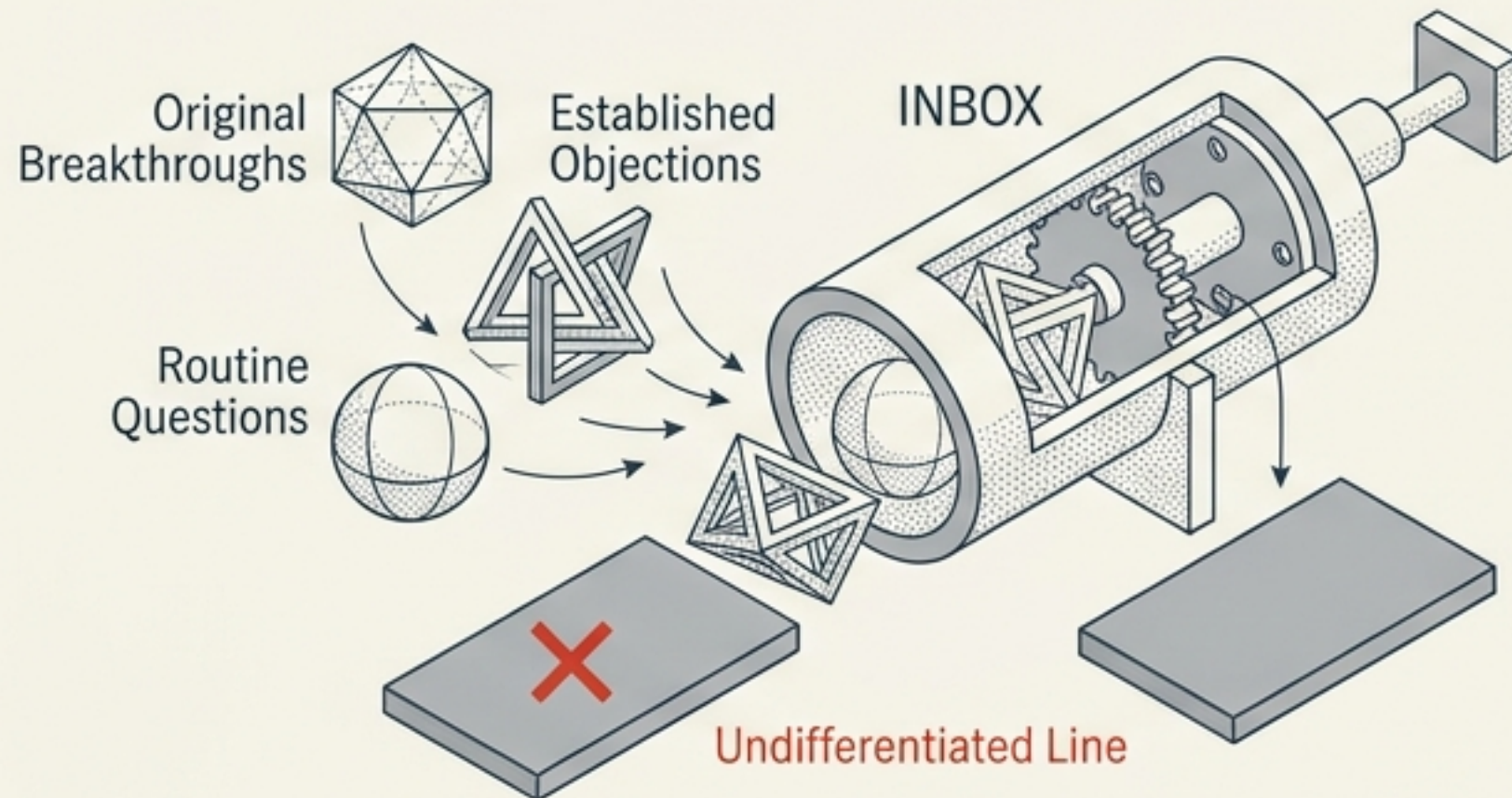


An architectural
blueprint for
many-to-one
intellectual
communication
under finite
recipient attention.



The Bottleneck: Digital communication scales access infinitely, but recipient attention ($\lambda E[C] < A$) is a strictly finite computational resource.

The Recolpction: The rates of trunking how about awiarning cognitive-rondoreda nturally r colaxing incoming communication and expected cognitive load $\times$ of A isa available attention.



The Inbox Failure: Chronological queues collapse routine questions, established objections, and original breakthroughs into one undifferentiated line.

The Consequence: The repeated loss of previously performed cognitive labor. The recipient is forced to reconstruct the same conceptual path from zero with every new arrival.

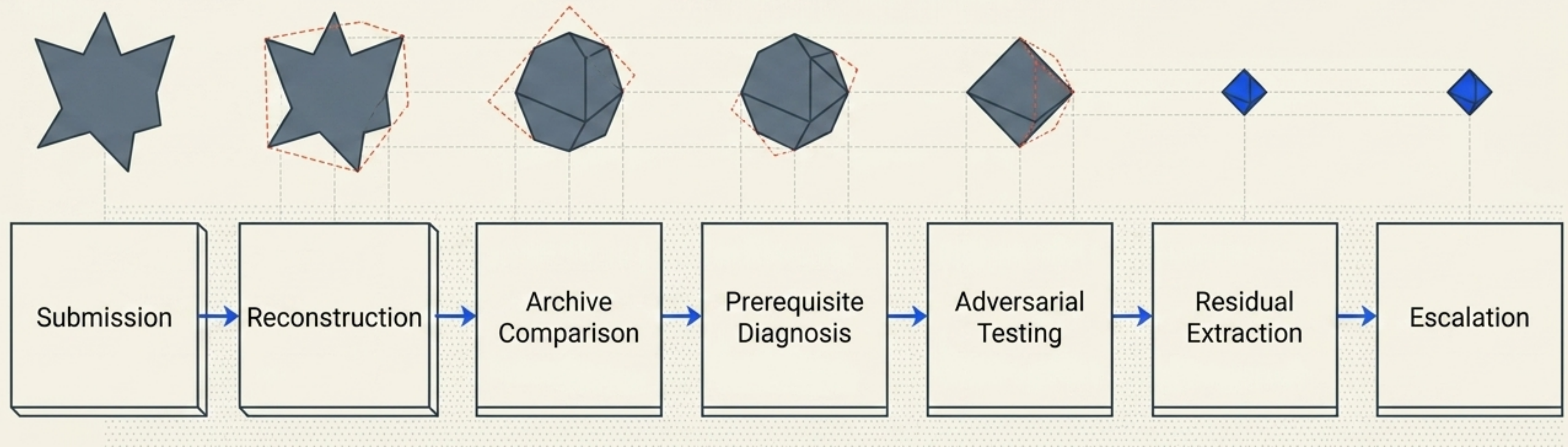
The Routing Paradigm Shift

Dimension of Comparison	Traditional Feed / Search	The Attention Compiler
State Model of User	Static demographic & keyword preference	Continuous, revisable epistemic state estimation (θ_i)
Treatment of Novelty	Amplifies surface-level linguistic unfamiliarity	Extracts semantic and formal residual novelty after charitable reconstruction
Handling of Duplicates	Redundant processing & chronological sorting	Dynamic routing through shared prerequisite nodes (collective recurrence)
Capacity Constraint	Solved by exclusionary gatekeeping or algorithmic suppression	Solved by transforming heterogeneous input into recipient-relative value

The Triage Pipeline: From Submission to Residual

“A compiler does not merely filter its input; it transforms expressions while detecting failures and preserving relevant structure.”

“The objective is not to immediately judge the value of a message, but to determine what mathematical and conceptual operations must be applied before its value can be assessed.”



Submissions relay into

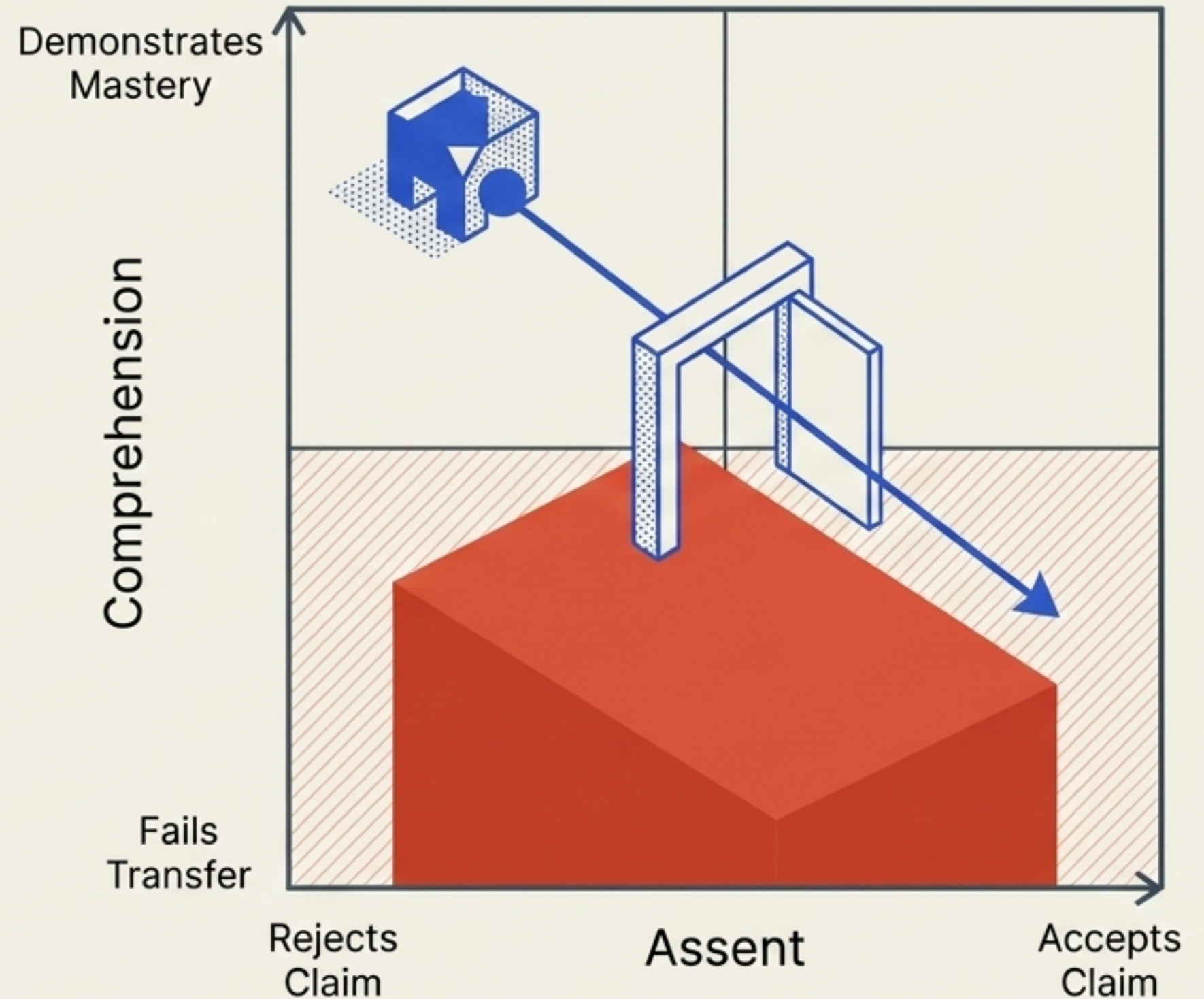
Information routing retains

Decoupling Comprehension from Assent

Routing depends on a provisional, continuously updated model of the interlocutor's epistemic state ($\theta_i(t)$).

Demonstration over Attendance: Users bypass prerequisites via successful application, explanation, or counterexample, not by clicking through ceremonial courses.

Ideological Neutrality: A participant passes a conceptual gate by accurately understanding a distinction, even if they reject its premise. Conflating rejection with incompetence turns prerequisite routing into ideological screening.



CLIO: Admissibility-Relative Projection

The Function:

CLIO compresses the raw submission (x) into a sufficient operational representation (m).

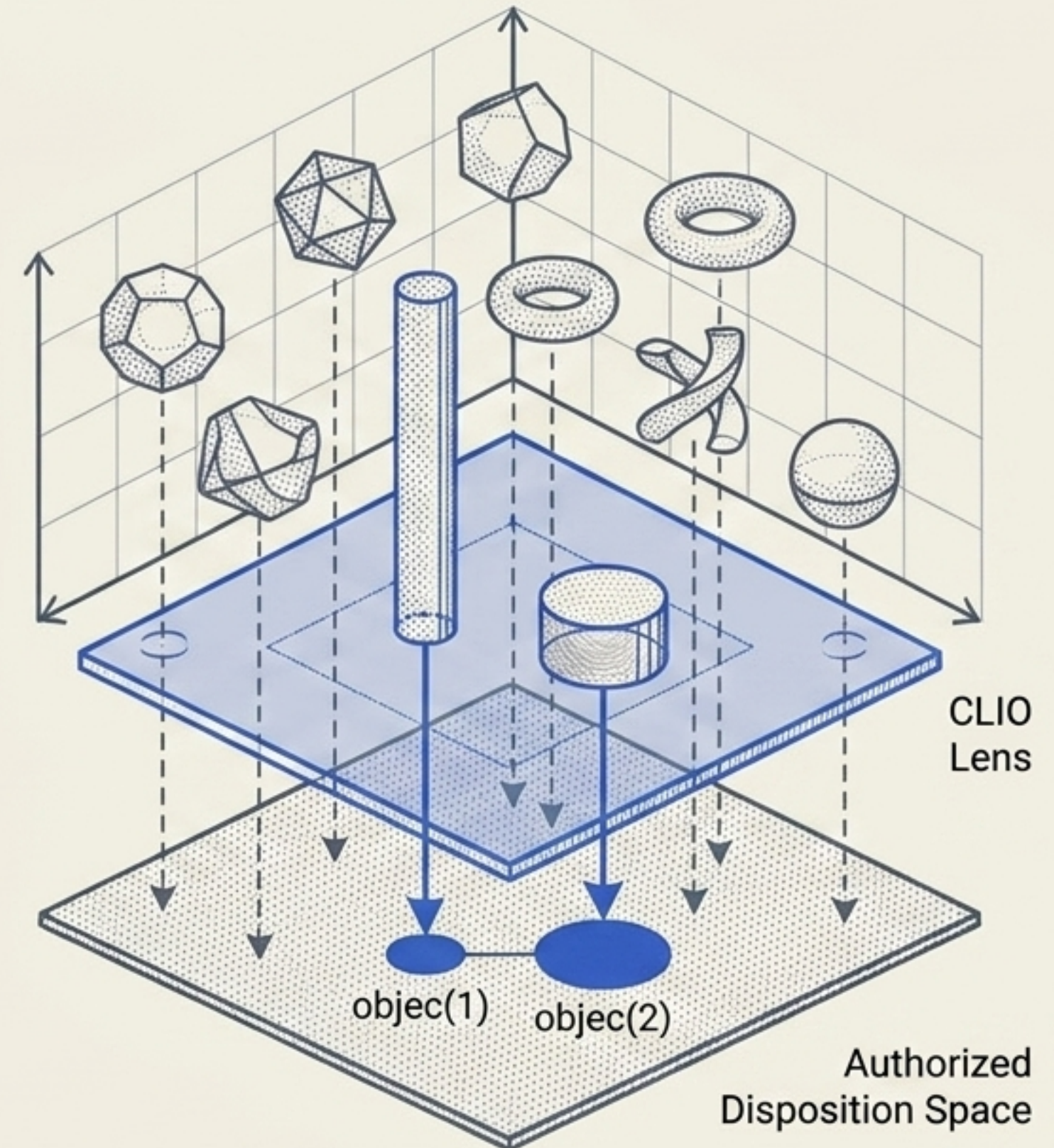
Representational Simplicity is Relative:

Removing a coordinate is only 'simple' if every authorized downstream action (escalation, audit, answer) is invariant under its removal.

The Axiom of Sufficiency:

$$q_c(x) = q_c(y) \implies f(x, c) = f(y, c).$$

If a distinction cannot alter any authorized action, retaining it is representational surplus.



The Danger of Projection Collapse

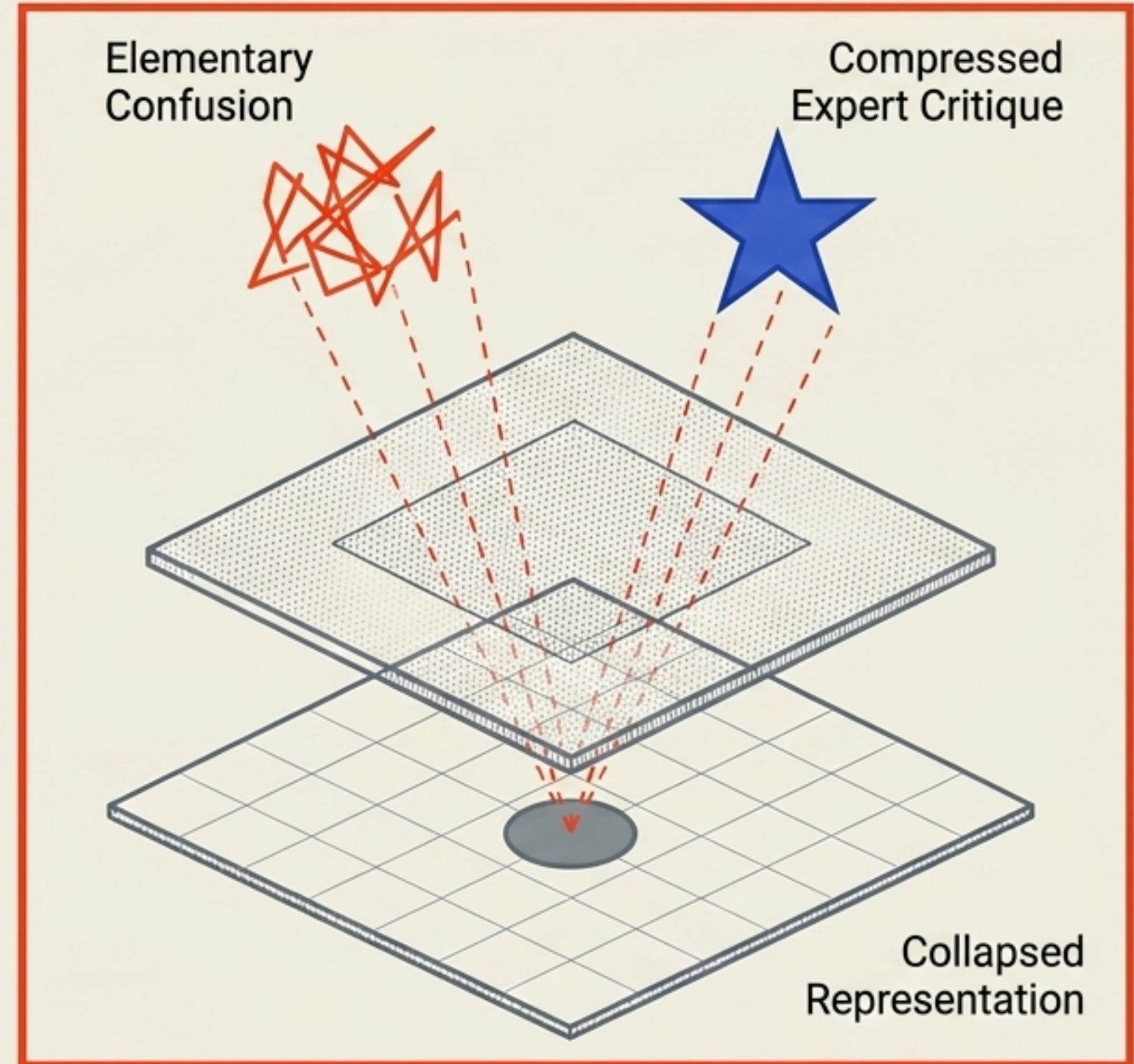
The Failure Mode:

Projection Collapse occurs when a quotient identifies submissions that require different authorized dispositions.

Minority-Value Collapse:

If an over-smoothed representation makes a highly valuable residual indistinguishable from a routine question, a cost-optimal system will suppress the novelty.

False suppression is often not a classification error, but a byproduct of rational cost-optimization over an insufficient representation.

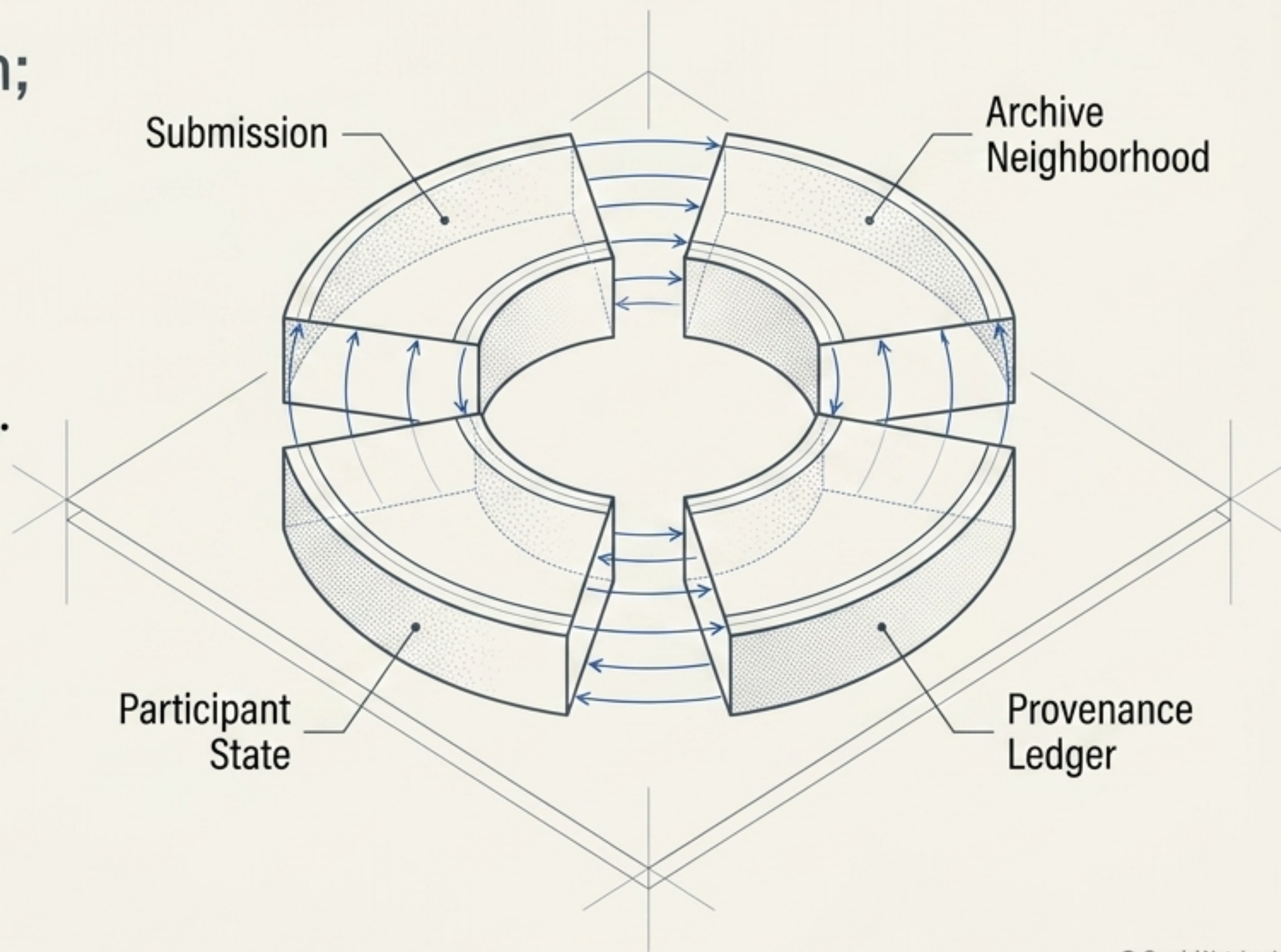


HYDRA: Contextual Assembly and the Colimit

CLIO compresses the submission;
HYDRA glues the context.

A routing decision depends simultaneously on the projected claim, demonstrated competencies, and recipient responsibilities.

Conversational Gluing:
HYDRA attempts to assemble these distributed local states into a coherent global conversational state via categorical colimits, constrained by explicit compatibility maps.



Gluing Obstructions and Cyclic Holonomy

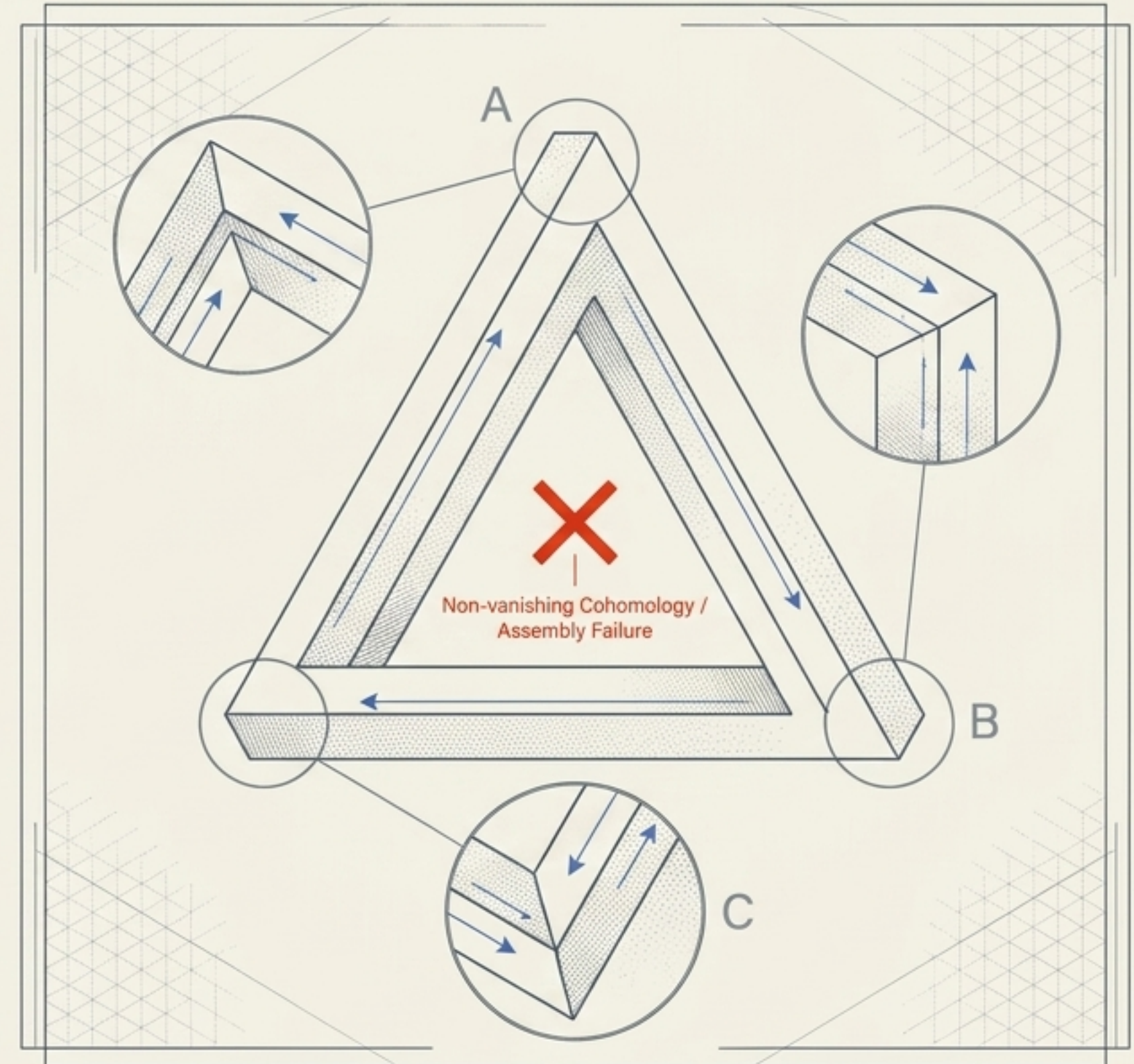
The Failure Mode:

Local coherence does not guarantee global coherence. A claim may align with the archive, and the archive with a prerequisite, but the full contextual loop contradicts itself.

The Obstruction Class:

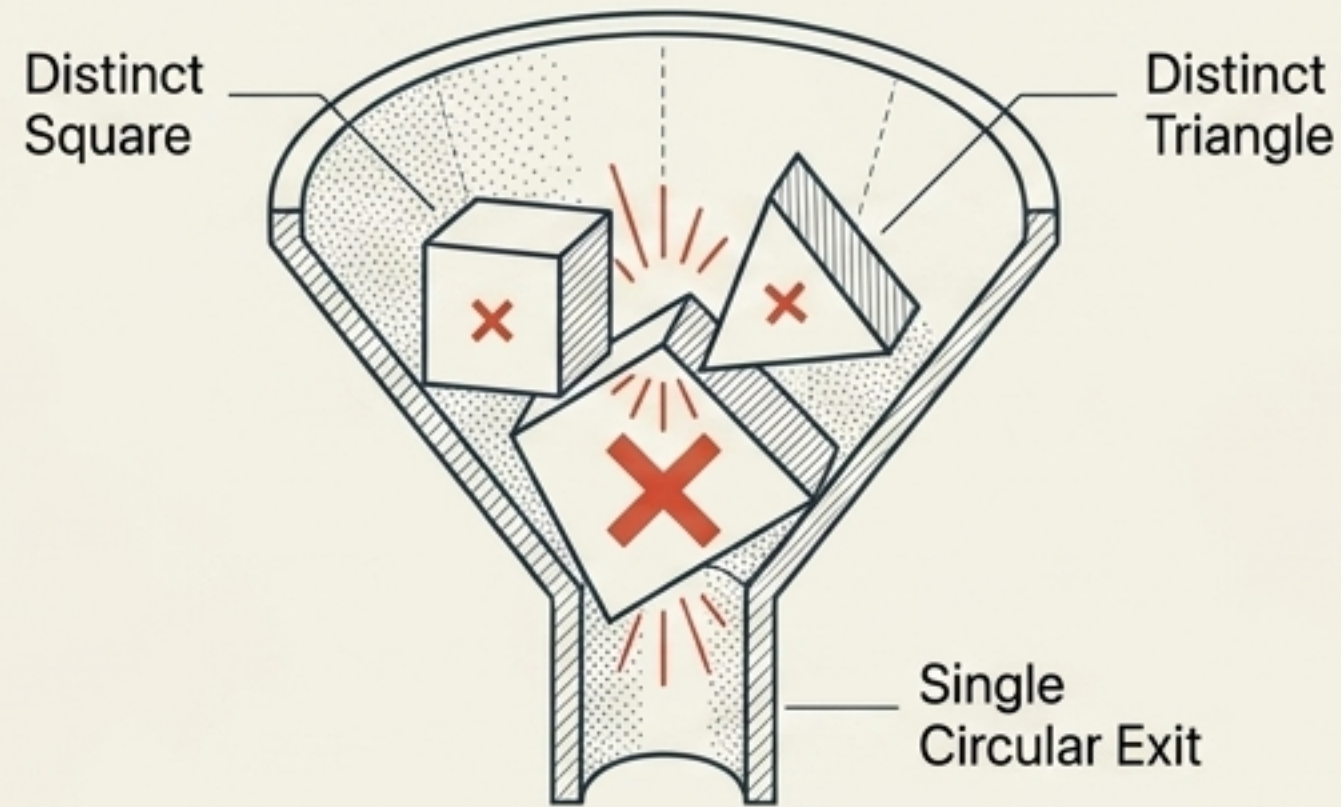
When local states contradict (e.g., incompatible provenances or conflicting translations), the system generates a non-vanishing gluing obstruction.

The system must report this architectural obstruction rather than forcing an invalid mathematical identification to maintain the illusion of a clean answer.



Diagnostic Matrix: Dual Failure Modes

CLIO: Projection Collapse



- Symptom: Over-identification. Treats distinct things as the same.
- Result: Suppresses novelty; false positives for routine answers.
- Repair: Expand features. Restore the discarded distinction to the quotient space.

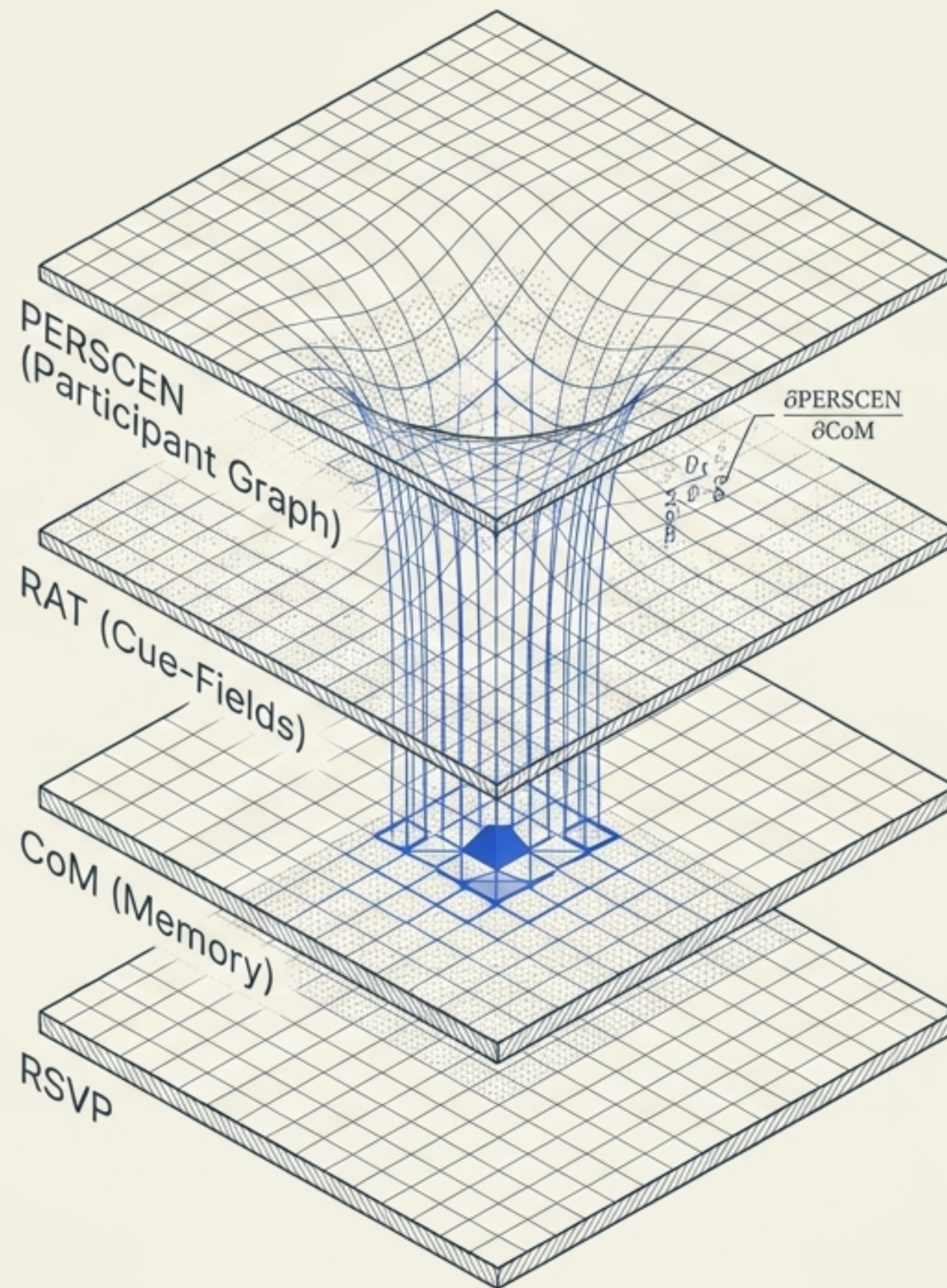
HYDRA: Gluing Obstruction



- Symptom: Fragmented State. Valid local truths form impossible global geometries.
- Result: System gridlock; contradictory context assignments.
- Repair: Branching state. Revise compatibility maps or escalate for human reconciliation.

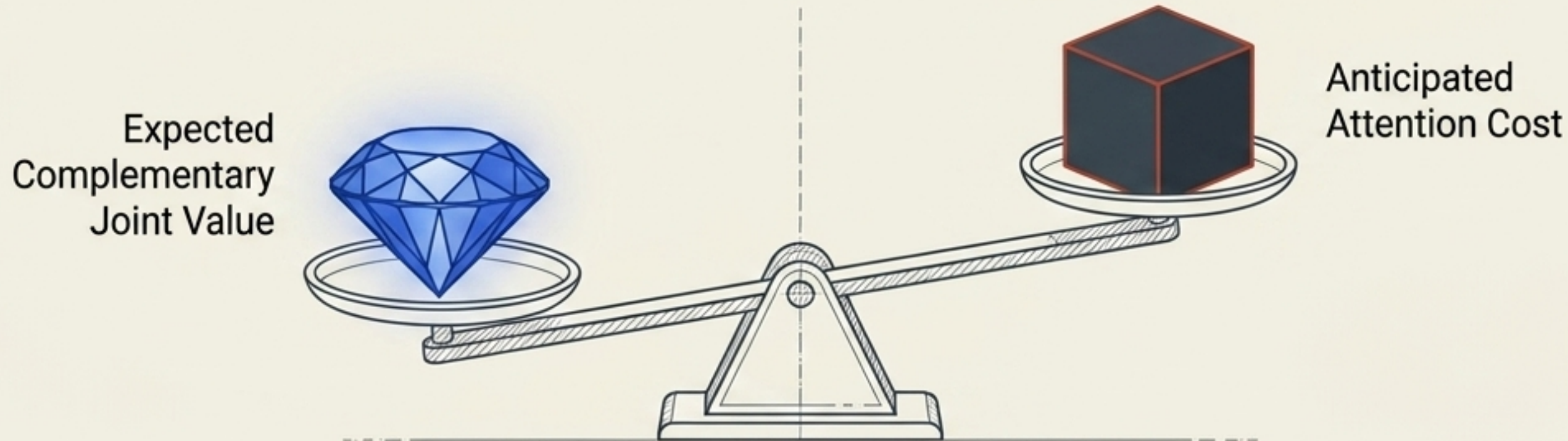
Parameter-Mediated Operational Coupling

Components in the Attention Compiler do not act in isolation. They are functionally nonseparable. A change in the Chain of Memory (CoM) alters the cue-density field (RAT), which actively reshapes the participant's competency graph (PERSCEN). True architecture must account for this parameter-mediated coupling via mixed-partial derivatives, proving that evaluation is not just an additive sum of isolated parts.



$$\mathbb{E}[\Delta_{joint}(x_i, r) \mid y, K] > c_i$$

The Escalation Function: Math of Scarce Attention



Escalation

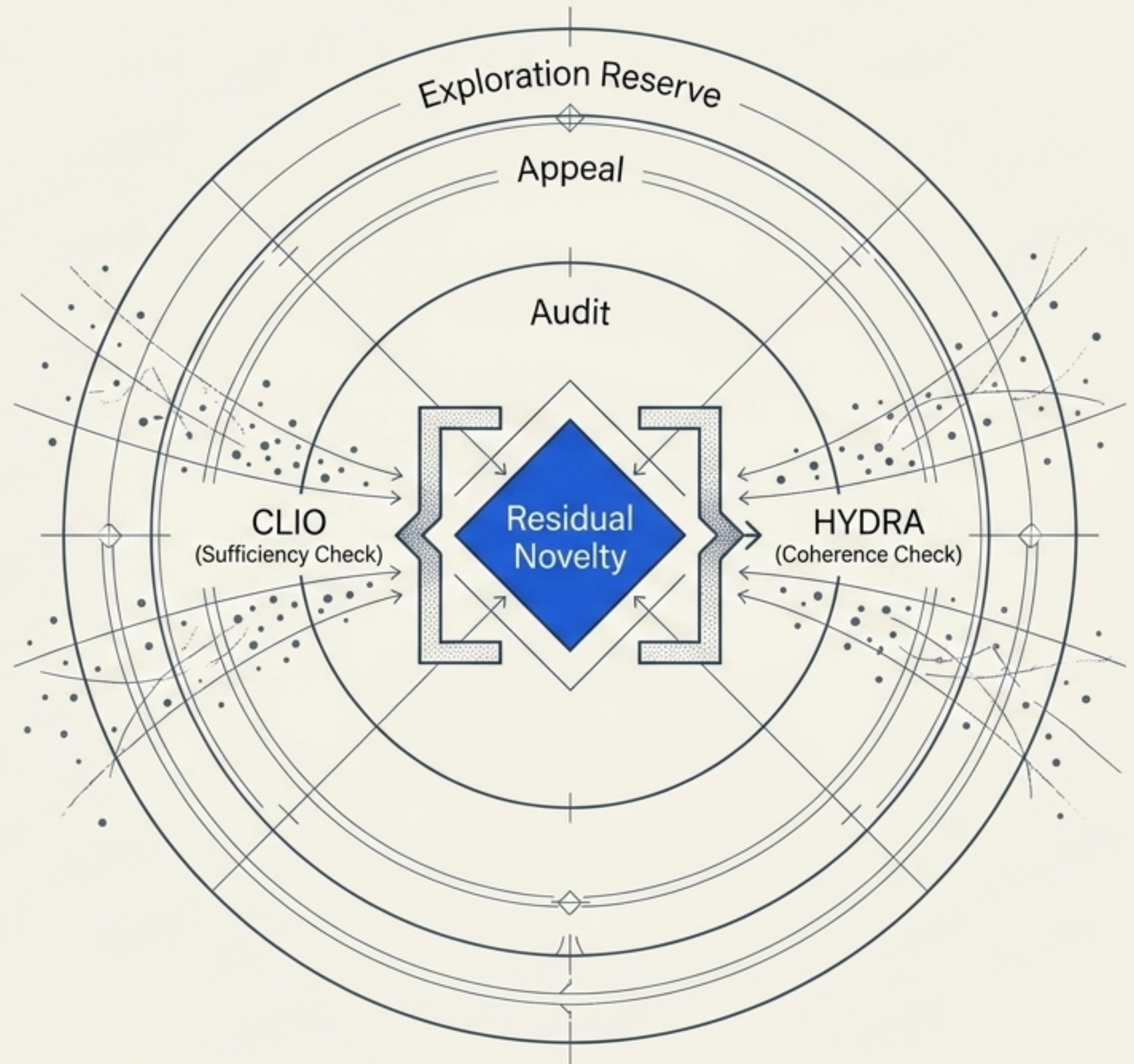
- Escalation is a resource-allocation decision, not a judgment of personal worth. It is strictly recipient-relative.
- The strongest reason for escalation is not independent novelty, but the complementary value generated between this specific participant and this specific recipient.

The Threshold

- A submission moves upward only when the expected intellectual gain from direct interaction exceeds the cost of the recipient's finite attention.

Synthesis: Dual Accountability

- Universal access and finite attention are reconciled through Transformative Access.
- Intellectual triage is only mathematically warranted if the representation is sufficient (CLIO) and the assembly is coherent (HYDRA).
- Governance safeguards prevent the convergence point from becoming a self-confirming gatekeeper.



The Frontier of the Conversation

The goal of the Attention Compiler is not exclusion.

It extracts the precise, unresolved remainder after answers, terminologies, prerequisites, and known objections are stripped away.

Every message must become more intelligible as it approaches the frontier of scarce attention, allowing the recipient to encounter the edge of the conversation rather than its preliminary history.

